Guidelines for XML-structure to import data in Smartschool Planner

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Examples of XML-structure

Basic example

```
<?xml version="1.0" encoding="utf-8"?>
<model xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
source="xxxxx">
        <elements>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c7c4d">
                        <type>lesson</type>
                        <time>
                                <start>
                                        <date>2022-03-11</date>
                                        <time>08:25:00</time>
                                </start>
                                <end>
                                        <date>2022-03-11</date>
                                        <time>09:15:00</time>
                                </end>
                        </time>
                        <classes>
                                <class>
                                        <code><![CDATA[5LWi]]></code>
                                </class>
                        </classes>
                        <teachers>
                                <teacher>
                                        <code><![CDATA[EMMKA]]></code>
                                </teacher>
                        </teachers>
                        <locations>
                                <location>
                                        <code><![CDATA[2.17]]></code>
                                </location>
                        </locations>
                        <courses>
                                <course>
                                        <code><![CDATA[WIS8]]></code>
                                </course>
                        </courses>
                </element>
        </elements>
</model>
```

Example of some different time combinations (all other tags are stripped for this example)

```
<?xml version="1.0" encoding="utf-8"?>
<model xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
source="xxxxx">
        <elements>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c7c4d">
                        <time>
                                <start>
                                        <date>2022-03-11</date>
                                        <time>08:25:00</time>
                                </start>
                                <end>
                                        <date>2022-03-11</date>
                                        <time>09:15:00</time>
                                </end>
                        </time>
                </element>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c8e7b">
                        <time>
                                <start>
                                        <day>5</day>
                                        <time>08:25:00</time>
                                </start>
                                <end>
                                        <day>5</day>
                                        <time>09:15:00</time>
                                </end>
                        </time>
                </element>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c9f6c">
                        <time>
                                <start>
                                        <date>2022-03-11</date>
                                        <time>08:25:00</time>
                                        <timezone>+01:00</timezone>
                                </start>
                                <end>
                                        <date>2022-03-11</date>
                                        <time>09:15:00</time>
                                        <timezone>+01:00</timezone>
                                </end>
                        </time>
                </element>
        </elements>
</model>
```

Example with multiple classes, teachers, locations and courses

```
<?xml version="1.0" encoding="utf-8"?>
<model xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
source="xxxxx">
        <elements>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c7c4d">
                        <type>lesson</type>
                        <time>
                                <start>
                                        <date>2022-03-11</date>
                                        <time>08:25:00</time>
                                </start>
                                <end>
                                        <date>2022-03-11</date>
                                        <time>09:15:00</time>
                                </end>
                        </time>
                        <classes>
                                <class>
                                        <code><![CDATA[2Aa]]></code>
                                </class>
                                <class>
                                        <code><![CDATA[2Ab]]></code>
                                </class>
                                <class>
                                        <code><![CDATA[2Ac]]></code>
                                </class>
                        </classes>
                        <teachers>
                                <teacher>
                                        <code><![CDATA[MARIV]]></code>
                                </teacher>
                                <teacher>
                                        <code><![CDATA[LIEKA]]></code>
                                </teacher>
                        </teachers>
                        <locations>
                                <location>
                                        <code><![CDATA[A3.11]]></code>
                                </location>
                                <location>
                                        <code><![CDATA[C8]]></code>
                                </locations>
                        </locations>
                        <courses>
                                <course>
                                        <code><![CDATA[AARD2]]></code>
                                </course>
                                <course>
                                        <code><![CDATA[FRANS3]]></code>
                                </course>
                        </courses>
                </element>
        </elements>
</model>
```

Example with class > partitions

```
<?xml version="1.0" encoding="utf-8"?>
<model xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
source="xxxxx">
        <elements>
                <element id="f29a132b-afd7-4582-a2f8-6caad48c7c4d">
                        <type>lesson</type>
                        <time>
                                <start>
                                        <date>2022-03-11</date>
                                        <time>08:25:00</time>
                                </start>
                                <end>
                                        <date>2022-03-11</date>
                                        <time>09:15:00</time>
                                </end>
                        </time>
                        <classes>
                                <class>
                                        <code><![CDATA[1A]]></code>
                                        <partitions>
                                                <partition>
                                                        <name><![CDATA[sex]]></name>
                                                        <values>
                                                                <value><![CDATA[male]]></value>
                                                        </values>
                                                </partition>
                                        </partitions>
                                </class>
                                <class>
                                        <code><![CDATA[1B]]></code>
                                        <partitions>
                                                <partition>
                                                        <name><![CDATA[sex]]></name>
                                                        <values>
                                                                <value><![CDATA[male]]></value>
                                                        </values>
                                                </partition>
                                        </partitions>
                                </class>
                                <class>
                                        <code><![CDATA[1C]]></code>
                                        <partitions>
                                                <partition>
                                                        <name><![CDATA[sex]]></name>
                                                        <values>
                                                                <value><![CDATA[male]]></value>
                                                        </values>
                                                </partition>
                                        </partitions>
                                </class>
                        </classes>
                        <teachers>
                                <teacher>
                                        <code><![CDATA[JANTO]]></code>
                                </teacher>
                        </teachers>
                        <locations>
                                <location>
                                        <code><![CDATA[Sporthal B2]]></code>
                                </location>
                        </locations>
                        <courses>
                                <course>
                                        <code><![CDATA[LOj]]></code>
                                </course>
                        </courses>
                </element>
        </elements>
</model>
```

Example with multiple partition values in the same class • Partitions in this example will be matched to a custom field in Smartschool

```
<?xml version="1.0" encoding="utf-8"?>
<model xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
source="xxxxx">
	<elements>
		<element id="f29a132b-afd7-4582-a2f8-6caad48c7c4d">
			<type>lesson</type>
			<time>
				<start>
					<date>2022-03-11</date>
					<time>08:25:00</time>
				</start>
				<end>
					<date>2022-03-11</date>
					<time>09:15:00</time>
				</end>
			</time>
			<classes>
				<class>
					<code><![CDATA[3ELEK]]></code>
					<partitions>
						<partition>
							<name><![CDATA[praktijk]]></name>
							<values>
								<value><![CDATA[groep 4]]></value>
								<value><![CDATA[groep 5]]></value>
							</values>
						</partition>
					</partitions>
				</class>
				<class>
					<code><![CDATA[4ELEK]]></code>
					<partitions>
						<partition>
							<name><![CDATA[praktijk]]></name>
							<values>
								<value><![CDATA[groep 4]]></value>
								<value><![CDATA[groep 5]]></value>
							</values>
						</partition>
					</partitions>
				</class>
			</classes>
			<teachers>
				<teacher>
					<code><![CDATA[PEEHE]]></code>
				</teacher>
			</teachers>
			<locations>
				<location>
					<code><![CDATA[Werkhuis E5.02]]></code>
				</location>
			</locations>
			<courses>
				<course>
					<code><![CDATA[ELEKpr]]></code>
				</course>
			</courses>
		</element>
	</elements>
</model>
```

Guidelines for every tag/attribute in the XML

Tag/attribute	Guidelines
source	Reference to the origin of the XML. • Required attribute • Value cannot be empty • Can only contain one value Contact our support team (desk@smartschool.be) if you want to be recognized as a source.

element id	Id of the element provided by the source. Smartschool uses this to match this element to an element in a previous import file from the same source. In that way elements across import files can be updated by a new import.
	
	For example: teacher JANTO has 4 hours of the course MATH with class 1A. This means there are 4 elements in one week. Across 5 weeks there are 20 elements.
	
	We are looking for 20 unique ids. Only then will the id be truly uniqe for an element and only then we can guarantee that the updating process of elements will be correct.
	
	If you're not able to provide a truly unique id (only 4 ids which are reused across the 5 weeks in the above example) then we can't guarantee that the updating process of elements will be correct. In this case we'll have to implement our own interpretation which will be **sensitive to errors**.
	
	• Required attribute
	• Value cannot be empty
	• Can contain only one value
	• Format:
	  • we prefer uuid4
	  • we are open to see what id you can provide

element type	Determines the type of the element. For now, there is only one type. This is a future proof tag. • Required tag • Value cannot be empty • Can only contain one value • Possible values: • lesson

element > time > start date
 	
Condition You can only use this tag if you provide a truly unique id (see '**element** id' above). If you're not able to provide a truly unique id, you're to use '**start** day' instead of '**start** date'.
	Determines the start date of the element.
	
	• Required tag, unless '**element > time > start** day' is present for this element
	• If tag is present:
	  • then value cannot be empty
	  • can only contain one value
	• Format: YYYY-MM-DD

element > time > start day
 	
We prefer the '**element > time > start** date' as this is more specific and more correct. This tag is only a back-up if '**element > time > start** date' or a truly unique '**element** id' is absolute not possible to return.
	Determines the day of the week on which the element starts.
	
	• Optional tag, becomes required when '**element > time > start** date' is not present for this element
	• If tag is required:
	  • then value cannot be empty
	  • can only contain one value
	• Possible values:
	  • 1 (corresponds with monday)
	  • 2 (corresponds with tuesday)
	  • 3 (corresponds with wednesday)
	  • 4 (corresponds with thursday)
	  • 5 (corresponds with friday)
	  • 6 (corresponds with saturday)
	  • 7 (corresponds with sunday)

element > time > start time	Determines the start time of the element. • Required tag • Value cannot be empty • Can only contain one value • Format: hh:mm:ss

element > time > start timezone	Determines the timezone of the start time of the element. This is an optional tag if the timezone from the source is not the same as the timezone of Belgium: Central European Time (UTC+01:00) and Central European Summer Time (UTC+02:00). For a Balgian source, this tag is optional. They can leave it out of the XML. • Optional tag • Can only contain one value • Format: +03:00 or -06:00 to determine the offset from the above mentioned time zone
element > time > end date	Same requirements as **element > time > start** date. Additional: • Required tag if '**element > time > start** date' is present.
element > time > end day	Same requirements as **element > time > start** day. Additional: • Required tag if '**element > time > start** day' has become required.
element > time > end time	Same requirements as **element > time > start** time. Additional: • Value must be at least one minute later then '**element > time > start** time' This is to ensure that an element lasts at least one minute.
element > time > end timezone	Same requirements as **element > time > start** timezone. Additional: • Required tag if '**element > time > start** timezone' is present and contains a value.

class code
 	
If your programming language struggles with 'class', you can also use 'classGroup'. We prefer you don't mix the usage of 'class' and 'classGroup', but make a choice between these two.
 	
classGroup code
	Determines which classes will be participating in this element.
	
	• Optional tag, becomes required when '**element** type' contains the value 'lesson' for this element
	• If tag is required:
	  • then value cannot be empty
	  • can contain multiple values
	• Format: <![CDATA[value]]> This is to ensure that all characters (e.g. &) can be processed
	
	Good to know for your customers:
	
	• The value of '**class** code' must match (case-insensitive) the value of '**koppelingsveld schoolagenda**' in Smartschool. This only applies in Smartschool to groups of the type officialClass and regularClass (not to the type regularGroup).
	• If a '**class** code' is not unique across all users in Smartschool, the element containing that code will not be processed.

class > partitions > partition **classGroup > partitions > partition**	A class can be divided into partitions. A partition contains a name and one or more values. This allows schools te refine their schedule. For example, physical education will be given by two teachers; one teacher for the boys and one teacher for the girls. Schools can create custom fields in Smartschool. A custom field in Smartschool contains a name and a type (radiobuttons, checkboxes). This allows schools to further refine their schedule.

class > partitions > partition name

classGroup > partitions > partition name

Determines which partitions exist within this class for this element.

- Optional tag
- Can contain multiple values. Multiple values will be processed as AND for this class in this element.
- Possible values:
  - 'geslacht' or 'sex' to match the 'sex' field that is present in Smartschool
  - 'godsdienst' or 'religion' to match the 'religion' field that is present in Smartschool
  - The value of '**class > partitions > partition** name' must match (case-insensitive) the name of a custom field in Smartschool
    - If a '**partition** name' is not unique across all custom fields values in Smartschool, the element containing that partition name will not be processed.

class > partitions > partition > values value

classGroup > partitions > partition > values value

Determines which partition values will be participating for this class in this element.

- Optional tag, becomes required when '**class > partitions > partition** name' is present for this class in this element
- Can contain multiple values. Multiple values will be processed as OR for this partition name in this class in this element.
- Possible values:
  - For the partition name 'sex' it must be one of the following values:

value in XML	meaning in Smartschool
jongen	Mannelijk male
male	Mannelijk male
meisje	Vrouwelijk female
female	Vrouwelijk female

  - For the partion name 'religion' it must be one of the following values:

value in XML	meaning in Smartschool
cul	Cultuurbeschouwing
cpd	Cultuurbeschouwing
ecr	Eigen cultuur en religie
per	Eigen cultuur en religie
isl	Islamitische godsdienst
isr	Israëlitische godsdienst
kgd	Katholieke godsdienst
cat	Katholieke godsdienst
ncz	Niet-confessionele zedenleer
mor	Niet-confessionele zedenleer
ort	Orthodoxe godsdienst
pgd	Protestantse godsdienst
pro	Protestantse godsdienst
ang	Anglicaanse godsdienst
vrg	Vrijgesteld of niet van toepassing
dis	Vrijgesteld of niet van toepassing

  - The value of '**class > partitions > partition** value' must match (case-insensitive) one of the values of the custom field in Smartschool that was matched to '**class > partitions > partition** name'
    - If a '**partition** value' is not unique across the values in the corresponding '**partition** name' in Smartschool, the element containing that partition value will not be processed.

teacher code	Determines which teachers will be organising this element. A teacher in Smartschool contains an id, name, surname and code. Since the combination of name and surname is not necessarily unique, we use the code to match teachers. • Required tag • Value cannot be empty • Can contain multiple values. Multiple values will be processed as AND for this element (co-teaching). • Format: <![CDATA[value]]> This is to ensure that all characters (e.g. &) can be processed Good to know for your customers: • The value of '**teacher** code' must match (case-insensitive) the value of '**koppeling sveld schoolagenda**' in Smartschool. This only applies in Smartschool to users with the role 'teacher', 'principal' of 'other' (not to the role 'student'). • If a '**teacher** code' is not unique across all users in Smartschool, the element containing that code will not be processed.
location code	Determines in which locations this element will take place. • Optional tag • Can contain multiple values. • Format: <![CDATA[value]]> This is to ensure that all characters (e.g. &) can be processed Good to know for your customers: • Schools have to create one or more locations databases in Smartschool. The value of '**location** code' must match (case-insensitive) the name of an item in one of those databases in Smartschool. • If a '**location** code' is not unique across those databases in Smartschool, the element containing that location will not be processed.
course code	Determines for which courses this element is organised. • Optional tag • Can contain multiple values. • Format: <![CDATA[value]]> This is to ensure that all characters (e.g. &) can be processed Good to know for your customers: • Schools have to create a list of courses in Smartschool. A course in that list contains a name and an external name. The value of '**course** code' must match (case-insensitive) the external name of a course in Smartschool. • If a '**course** code' is not unique across all courses in this list in Smartschool, the element containing that course will not be processed.

Good to know

- When importing data via this XML, Smartschool performs a check. Are there elements that don't comply with the above guidelines? Then the user will be given a simply error report. The user then decides (A) or (B). This ensures that the user doesn't have to edit the XML to delete/edit any unnecessary data.
    - (A) Abort the process and import nothing
    - (B) Continue the process and import all the compliant elements. The non-compliant elements will not be imported.